

Rye Bread Consumption in Early Life and Reduced Risk of Advanced Prostate Cancer

Objective To determine whether consumption of whole-grain; rye bread, oatmeal, and whole-wheat bread, during different periods of life, is associated with risk of prostate cancer (PCa). **Methods** In 2002 to 2006, 2,268 men, aged 67-96 years, reported their dietary habits in the AGES-Reykjavik cohort study. Dietary habits were assessed for early-, mid-, and current life using a validated food frequency questionnaire (FFQ). Through linkage to cancer- and mortality registers, we retrieved information on PCa diagnosis and mortality through 2009. We used regression models to estimate odds ratios (ORs) and hazard ratios (HRs) for PCa according to whole grain consumption, adjusted for possible confounding factors including fish-, fish liver oil-, meat-, and milk intake. **Results** Of the 2,268 men, 347 had or were diagnosed with PCa during follow-up, 63 with advanced disease (stage 3+ or died of PCa). Daily rye bread consumption in adolescence (vs. less than daily) was associated with a decreased risk of PCa diagnosis (OR = 0.76, 95% Confidence interval (CI): 0.59-0.98), and of advanced PCa (OR = 0.47, 95% CI: 0.27-0.84). High intake of oatmeal in adolescence (≥ 5 vs. ≤ 4 times/ week) was not significantly associated with risk of PCa diagnosis (OR = 0.99, 95% CI: 0.77-1.27) nor advanced PCa (OR = 0.67, 95% CI: 0.37-1.20). Mid-, and late life consumption of rye bread, oatmeal, or whole-wheat bread was not associated with PCa risk. **Conclusion** Our results suggest that rye bread consumption in adolescence may be associated with reduced risk of PCa, particularly advanced disease.